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Hazard of novel anti-fouling nanomaterials and biocides DCOIT and silver to marine organisms†

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Biocide-based coatings have been applied for several years to tackle marine biofouling. Recently, engineered nanomaterials, such as silica mesoporous nanocapsules (SiNCs), were used to encapsulate DCOIT, an anti-fouling booster biocide, and Ag, a broad-spectrum bactericidal agent, in order to control their release with time. This study aims to assess the environmental hazard of three novel anti-fouling nanomaterials (SiNC–DCOIT, SiNC–Ag and SiNC–DCOIT–Ag) and their free counterparts (DCOIT, Ag and SiNCs) to marine ecosystems. In order to accomplish this goal, marine toxicity data (L/E/IC₅₀ or NOEC) were compiled to derive PNEC values for each compound based on statistical and deterministic approaches. The first approach relies on the hazardous concentration for 5% of the species (HC₅ using species sensitivity distribution curves) and the second relies on the lowest L/E/IC₅₀ or NOEC value, considering the appropriate assessment factors. In both approaches, PNEC values were lower for the encapsulated biocides than for their free forms. Thus, biocide encapsulation appears to be a promising solution to develop a new generation of efficient anti-fouling additives for maritime coatings with lower environmental hazard than the current state-of-the-art biocides.

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Environmental significance

Biofouling in man-made immersed structures is an unsolved problem that causes severe economic and environmental impacts. Recently, it was demonstrated that some anti-fouling additives widely used in maritime coatings could be less toxic towards several marine organisms when immobilized in smart engineered nanomaterials compared to their respective conventional “non-nano” forms. The present study provides the first environmental hazard assessment of novel anti-fouling nanomaterials (SiNC–Ag, SiNC–DCOIT, SiNC–DCOIT–Ag) to marine ecosystems. Statistical and deterministic approaches were used to derive the most conservative hazardous endpoints for each compound (both nano- and non-nano forms). Biocide encapsulation appears to be a promising solution to develop a new generation of efficient anti-fouling additives for maritime coatings with lower environmental hazard than the current state-of-the-art biocides.

1. Introduction

Organotin substances (particularly TBT) have been widely used as anti-fouling agents since the late 1950s.^{1–3} However, these chemicals were banned in 2008 (ref. 4) and many new biocides have been developed and used in commercial anti-fouling (AF) paints to replace them,^{1,5,6} such as 4,5-dichloro-2-octyl-2H-isothiazol-3-one, known as DCOIT (CAS no.: 64359-81-5), a large spectrum booster biocide against soft and hard fouling.⁷ This active substance is easily biodegraded with a half-life of less than 24 hours in natural seawater.^{5,8,9} Although presenting rapid degradation in seawater, this biocide still poses risk to the environment as it has a broad-

spectrum action and can affect non-target organisms from lower to higher trophic levels in the first hours of activity, affecting the marine food chain.^{1,6,10} Besides this, when applied in paints, its release can be continuous leading to a potential prolonged exposure. Alternative techniques to prevent biofouling have also been developed, such as the use of enzymes, biomimetization, electrolysis of seawater, vibration, among others,^{3,11–15} but they are still not able to substitute biocide-based AF coatings. The use of biocides continues to be considered the most efficient, resistant and easiest technique, with low maintenance and cost-effectiveness. On the other hand, there is an increasing pressure from regulators and industry to develop new solutions with high AF efficacy and reduced or null toxicity for the remaining marine biota.¹⁶ In this context, nanotechnology has been recently proposed as an alternative eco-friendly solution based on AF biocide encapsulation or immobilization.^{17–23} Basically, the assumption of this technique relies on the controlled release of the biocides to

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decrease the undesired early lixiviation and all related environmental and economic impacts. DCOIT, zinc pyrithione and copper pyrithione were encapsulated in silica mesoporous nanocapsules (SiNCs)^{17,18} and/or immobilized in inorganic nanoclays.¹⁸ In order to enhance the anti-fouling efficacy of SiNC–DCOIT, Smallmatek Lda., the company responsible for the development and/or industrial validation of most above-mentioned AF nanostructured additives, proposed an external coating with silver nitrate (SiNC–DCOIT–Ag). The assumption is that the microbial biofilm formation is one of the first and critical stages of biofouling and silver is known as a bactericidal agent that kills a broad-spectrum of microorganisms.^{24–26} The environmental behavior, toxicity and anti-fouling efficacy of SiNC–DCOIT, SiNC–Ag and SiNC–DCOIT–Ag and their respective counterparts in target and non-target organisms were recently

reported,^{17,21} however their potential ecological hazard was not yet studied. A strict environmental regulation (Biocidal Products Regulation (BPR) EU No. 528/2012, REACH, CLP) has to be followed and accomplished to approve the use and commercialization of anti-fouling biocide-based substances in the European market. The BPR advises to carry out a dose–response approach to predict concentrations where no adverse effects will occur.

The predicted no-effect concentrations (PNECs) of a substance should be then derived from the available ecotoxicity testing data with adequate assessment factors. The PNEC is usually derived by a deterministic approach using the lowest NOEC (no observed effect concentration) value in a dataset, with an adequate safety factor. The other alternative approach to predict the PNEC value includes a statistical extrapolation methodology based on species

Table 1 Hazardous concentration for 5% of the species (HC₅) and predicted no effect concentrations (PNECs) (μg L⁻¹) for different biocides and nanomaterials obtained from the SSD curves constructed with different species toxicity data. n.a. – no information available; ST chronic – short-term chronic; DCOIT – 4,5-dichloro-2-octyl-4-isothiazolin-3-one; PTPB – pyridine-triphenylborane; TBTO – tributyltin oxide; MWCNT – multiwall carbon nanotubes; CB – carbon black; nano-nZVI: nano zerovalent iron

Type of substance	Environmental compartment	Substance	Unit	Parameter	HC ₅ (μg L ⁻¹)	PNEC _{stat} (μg L ⁻¹)	PNEC _{determ} (μg L ⁻¹)	Ref.
Biocides	Marine	DCOIT	μg DCOIT per L	Acute L/E/IC ₅₀	0.5	0.1 ^a	0.0004	Mochida <i>et al.</i> ³⁶
		PTPB	μg PTPB per L	Acute L/E/IC ₅₀	0.8	0.16 ^a	0.03	Mochida <i>et al.</i> ³⁵
		TBTO	μg TBT per L	Chronic L/NOECs	0.004	0.008 ^a	n.a.	Leung <i>et al.</i> ³¹
Nanomaterials	Freshwater	Silver	μg Ag ⁺ per L	Acute L/E/IC ₅₀	0.5	0.1 ^a	n.a.	CCME ⁴²
		TBTO	μg TBT per L	Chronic L/NOECs	0.03	0.006 ^a	n.a.	Leung <i>et al.</i> ³¹
		Silver	μg Ag ⁺ per L	Acute L/E/IC ₅₀	8	1.6 ^a	n.a.	CCME ⁴²
		MWCNT	μg MWCNT per L	n.a.	n.a.	n.a.	43	Lützhøf <i>et al.</i> ⁸⁰
	Marine	CB	μg CB per L	n.a.	n.a.	n.a.	5000	Lützhøf <i>et al.</i> ⁸⁰
		Nano-CuO	μg Cu per L	Acute L/EC ₅₀	150	30 ^a	n.a.	Adam <i>et al.</i> ⁸¹
				Acute LC ₅₀	n.a.	n.a.	0.34	Lützhøf <i>et al.</i> ⁸⁰
		Nano-Au	μg Au per L	Acute L/E/C ₅₀	42 780	8556 ^a	n.a.	Botha <i>et al.</i> ⁸²
				Acute/chronic NOEC	0.02	0.02	n.a.	Coll <i>et al.</i> ⁸³
		Nano-TiO ₂	μg Ti per L	Acute EC ₅₀	n.a.	n.a.	0.012	Lützhøf <i>et al.</i> ⁸⁰
				Acute/chronic NOEC	16	16	n.a.	Coll <i>et al.</i> ⁸³
		Nano-ZnO	μg Zn per L	ST chronic NOEC	n.a.	n.a.	18	Lützhøf <i>et al.</i> ⁸⁰
				Acute L/EC ₅₀	60	12 ^a	n.a.	Adam <i>et al.</i> ⁸¹
	Freshwater	Carbon nanotubes (CNTs)	μg CNTs per L	Acute/chronic NOEC	1	1	n.a.	Coll <i>et al.</i> ⁸³
				ST chronic NOEC	n.a.	n.a.	2.5	Lützhøf <i>et al.</i> ⁸⁰
		Fullerenes	μg fullerenes per L	Acute/chronic NOEC	56	56	n.a.	Coll <i>et al.</i> ⁸³
				ST chronic NOEC	4	4	n.a.	Lützhøf <i>et al.</i> ⁸⁰
		Nano-nZVI	μg n-nZVI per L	Acute LOEC	n.a.	n.a.	5	Coll <i>et al.</i> ⁸³
		Nano-CeO ₂	μg n-CeO ₂ per L	Acute LC ₅₀	n.a.	n.a.	5.2	Lützhøf <i>et al.</i> ⁸⁰

^a PNEC_{stat} derived from the present study using an assessment factor of 5 (AF = 5).

sensitivity distribution (SSD) curves from which the HC_5 value is derived, the hazardous concentration for 5% of the species.^{23–28} Although the use of chronic toxicity data is the most appropriate for constructing²⁸ the SSDs, where the magnitude and types of effects are ecologically more relevant and protection is higher, and acute toxicity data are more used due to high data availability. Additionally, the biocide hazard assessment proposed in the present study also reflects rapid degradation of DCOIT in seawater. SSDs have been developed for a wide variety of organic and inorganic chemicals, most of them in freshwater ecosystems^{27,29} due to the abundance of reliable data for this compartment, particularly in the case of nanomaterials^{30–33} (Table 1). Regarding marine species, there are SSDs for metals,^{32,33} pesticides³⁰ and anti-fouling substances (tributyltin,^{31,34} pyridine triphenylborane,³⁵ DCOIT³⁶). Although there are marine PNEC values for multi-walled carbon nanotubes and carbon black (Table 1), no SSDs for nanomaterials with marine species exist, as far as we are concerned.

Therefore, considering the amount of data generated for different novel anti-fouling nanomaterials²¹ and the data available in the literature, the main goal of this study was to predict the PNEC values for single substances SiNC, DCOIT and Ag and novel nanomaterials containing one or two biocides (SiNC–DCOIT, SiNC–Ag and SiNC–DCOIT–Ag), through a statistical and deterministic approach.

2. Experimental

2.1. Data selection

$L/E/IC_{50}$ and/or NOEC values for all compounds were retrieved from the literature, particularly from Figueiredo *et al.*²¹ in the case of the nanomaterials. In addition, ecotoxicological tests with the fish *Sparus aurata* (see section 2.2) were performed to supplement the data for SiNC, DCOIT, SiNC–DCOIT and SiNC–DCOIT–Ag (*cf.* Table S1† for new and compiled $L/E/IC_{50}$ data and Table S2† for NOEC data). These data were used to construct species sensitivity distribution curves and to calculate the HC_5 values and derive the marine PNEC values for each substance. The data tables were compiled using the following references: ref. 19, 39, 44, 46, 51–79.

2.2. Ecotoxicity testing

The acute toxicity testing with the fish *Solea senegalensis* followed the guideline OECD 236 (ref. 37) with adaptations. The 24 h exposure tests were performed in 24 well plates, and for each substance, five (SiNC: 333; 1000; 3330; 10 000; 33 330 μg SiNC per L; SiNC–DCOIT: 100; 300; 900; 2700; 8100 μg DCOIT per L) or six (SiNC–DCOIT–Ag: 3; 10; 90; 270; 810 μg DCOIT per L or 2.8; 9; 28; 83; 250; 750 μg Ag^+ per L) exposure concentrations, plus a negative control (artificial saltwater, salinity: 35 ± 1) were tested. A total of 20 replicates were tested per treatment. Each replicate contained 2 mL of the substance tested and 1 egg. The endpoint was the hatching rate and the tests were performed at a temperature

of 19 ± 1 °C and a photoperiod of 16 h : 8 h (light : dark). pH and dissolved oxygen were monitored for all treatments at the beginning. The medium was not replaced throughout the test.

2.3. EC_{50} and NOEC determination

The EC_{50} values for each substance were calculated with the software GraphPad Prism v6.0 using the non-linear regression equation that best fits to the data, considering the R^2 value, the absolute sum of squares and the 95% confidence intervals.

The NOEC values from the tests carried out in this study (section 2.2) were derived from one-way analysis of variance (one-way ANOVA) using the software SigmaPlot v12.5.

Data normality and homoscedasticity were tested with the Shapiro–Wilks and Bartlett tests, respectively. If these assumptions were valid, a one-way ANOVA test was performed to evaluate significant differences between treatments and control ($p < 0.05$), followed by the multiple comparison Dunnett test. If the data did not satisfy the normality or the homoscedastic assumptions, data transformation (*i.e.*, square, Ln, log10, reciprocal, exponential and square root) was carried out. If these transformations did not meet the normality and homoscedasticity assumptions, a one-way ANOVA on ranks was performed followed by Dunn's test.

2.4. Statistic PNEC determination

The species sensitivity distributions (SSDs) and the hazardous concentration at 5% (HC_5) were estimated using a log-normal distribution through the software ETX 2.1, based on the $L/E/IC_{50}$ and NOEC values. The geometric mean of the values was calculated when more than one value existed for the same species.

Following the technical guidance document,³⁸ the derivation of the statistic PNEC was calculated through the quotient between the HC_5 and the most conservative assessment factor,⁵ due to the use of acute and short-term chronic data.

2.5. Deterministic PNEC determination

The PNEC was determined using the lowest NOEC values for the dataset available for each test substance. This data set included the NOECs derived from this study (see above) and from Figueiredo *et al.*²¹ In addition, another deterministic exercise was carried out to determine the PNEC based on the lowest $L/E/IC_{50}$ available. The assessment factors used for each approach are shown in Table 2. Since the $L/E/IC_{50}$ dataset for all the substances contained short-term data for at least three taxonomic groups of three trophic levels, plus additional marine taxonomic groups, an assessment factor of 1000 was applied to calculate the deterministic PNEC for all substances, following the technical guidance document.³⁸ Regarding the approach based on NOEC values, an assessment factor of 100 was used to calculate the

Table 2 Assessment factors used for each substance proposed for deriving PNEC_{deterministic} for saltwater for different data sets (TGD, 2003)

Substances	Approach	Assessment factor	Data set
All	L/E/IC ₅₀ data	1000	Lowest short-term L/E/IC ₅₀ from freshwater or saltwater representatives of three taxonomic groups of three trophic levels + two additional marine taxonomic groups
All	NOEC data	100	Lowest long-term NOECs from three freshwater or saltwater species three trophic levels ^a

^a Reduced assessment factor (to 100, if only one short-term test or to 50, if two short-term tests on marine species are available) may be applied to the lowest NOEC from only two species when short-term tests for additional species representing marine taxonomic groups (e.g., echinoderms or molluscs), were carried out.

deterministic PNEC values, since the dataset contained NOECs for microalgae and crustacean and an additional short-term NOEC for echinoderms and/or fish.

3. Results

The hatching success data regarding the fish embryotoxicity test are summarized in Tables S1 and S2† together with the compiled literature data. In the beginning, the average pH and dissolved oxygen values were 8.00 ± 0.02 and $80 \pm 1\%$, respectively, and at the end of the test the average values of pH and dissolved oxygen were 7.94 ± 0.04 ; $56 \pm 1\%$, respectively.

The SSDs constructed with the L/E/IC₅₀ data are shown in Fig. 1(A to G) and their respective estimated HC₅ values in Table 3. Taking into consideration the content of the active substances, the lowest HC₅ value was observed for the free DCOIT (1 µg DCOIT per L) and the highest was for SiNC–DCOIT (26 µg DCOIT per L). Moreover, comparing the encapsulated forms of DCOIT and Ag, the value of HC₅ is higher for the single encapsulated forms (2.3-fold and 2.4-fold higher in the forms SiNC–DCOIT and SiNC–Ag, respectively) than for the dual encapsulated form (SiNC–DCOIT–Ag). Regarding SiNC–DCOIT–Ag, a HC₅ value 11-fold higher than for the free form was calculated for DCOIT, however, in terms of the silver content, the HC₅ of the encapsulated substance is almost the same as the free Ag form. The analysis of the SSD curves demonstrated that the most sensitive species to SiNC, DCOIT, Ag, SiNC–DCOIT, SiNC–DCOIT–Ag and SiNC–Ag exposure were the bryozoan *Bugula neritina* (Fig. 1A), the microalgae *Emiliana huxleyi* (Fig. 1B), the dinoflagellate *Lingulodinium polyedrum* (Fig. 1C), the diatom *Phaeodactylum tricornutum* (Fig. 1D and G) and the sea-urchin *Paracentrotus lividus* (Fig. 1E). As a function of the active compound concentration, the highest PNEC value was recorded for SiNC–DCOIT and the lowest for the free DCOIT (Table 3).

Fig. 2(A to G) show the SSDs based on the NOEC values. The estimated HC₅ values are shown in Table 4. In terms of active substance concentration, the lowest HC₅ value was observed for free DCOIT and the highest was for SiNC–Ag (as Ag content).

The HC₅ values for DCOIT calculated for encapsulated and free forms were considered similar in the case of SiNC–

DCOIT and 12-fold higher for SiNC–DCOIT–Ag. Regarding ionic silver, the HC₅ value was up to 2-fold higher for both encapsulated forms compared to free Ag⁺. Unlike the curves constructed with the L/E/IC₅₀ data, the HC₅ value for SiNC–DCOIT was 12-fold lower than SiNC–DCOIT–Ag (expressed in terms of the content of both biocides). For the encapsulated substances, the highest HC₅ value was found for SiNC–Ag, followed by SiNC–DCOIT–Ag and SiNC–DCOIT (cf. Table 4).

The diatom *P. tricornutum* was the most sensitive species to SiNC exposure (Fig. 2A). The microalgae *Skeletonema costatum* and *Isochrysis galbana* were the most sensitive species to DCOIT and Ag⁺ exposures, respectively (Fig. 2B and C). The most sensitive species to SiNC–DCOIT, SiNC–Ag and SiNC–DCOIT–Ag were the microalgae *P. tricornutum*, *I. galbana* and *N. gaditana*, respectively (Fig. 2D–G). SiNC–Ag showed the highest PNEC values (both approaches) followed by SiNC–DCOIT–Ag and SiNC–DCOIT; DCOIT has the lowest PNEC values from all tested substances (Table 4).

4. Discussion

The hazard assessment of anti-fouling biocides in the marine environment is scarcely known (Table 1) despite their deleterious effects on non-target and fouler species. There is only the study by Mochida *et al.*³⁶ where the HC₅ of DCOIT was estimated for marine organisms as 0.5 µg DCOIT per L (cf. Table 1), based on the acute L/E/IC₅₀ data. This DCOIT HC₅ value is half of the value estimated in the present study. This highlights the importance of collecting data from representative marine species of different trophic levels, to refine and to increase the reliability of the obtained estimates. Mochida *et al.*³⁵ estimated a lower HC₅ value for pyridine triphenylborane (PTPB) (Table 1) compared with DCOIT in this study. Regarding other anti-fouling biocides, Leung *et al.*³¹ demonstrated the extreme hazard of TBTO towards marine organisms, 250 times higher than DCOIT, comparing the chronic N/LOEC values (cf. Table 1) with the acute and short-term chronic NOEC values of DCOIT in the present study.

Comparing the obtained deterministic PNEC values with the values collected from the literature, the estimated PNEC for DCOIT in this study is similar to that estimated by Mochida *et al.*³⁶ and 1.4-fold lower than the PNEC presented

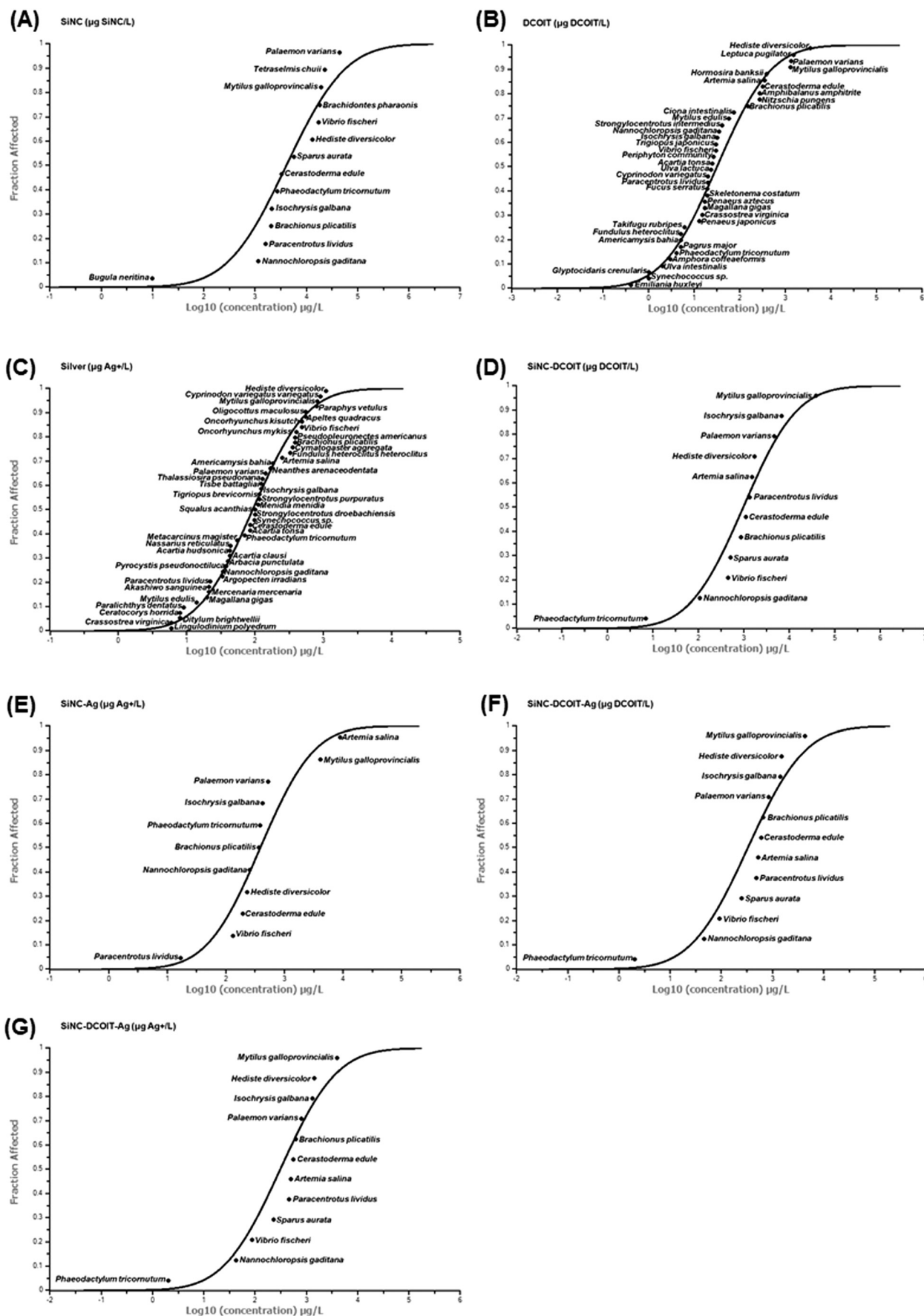


Fig. 1 Species sensitivity distributions for: (A) SiNC; (B) DCOIT; (C) Ag⁺; (D) SiNC-DCOIT (as DCOIT content); (E) SiNC-Ag (as Ag⁺ content); (F) SiNC-DCOIT-Ag (as DCOIT content); (G) SiNC-DCOIT-Ag (as Ag⁺ content) using the L/E/IC50 data for different species.

Table 3 Hazardous concentrations at 5% (HC₅) obtained from species sensitivity distributions using the L/E/IC₅₀ data for the different species and derived PNEC values. *N* is the sample size and the tests for normality included the Anderson–Darling test, the Kolmogorov–Smirnov test and the Cramér–von Mises test. PNEC_{stat} – derived from the HC₅ with AF = 5; PNEC_{determ} – derived from the lowest L/E/IC₅₀ with AF = 1000 (according to TGD (2003))

Contaminant	Units	<i>N</i>	HC ₅	CI 95%	Normality tests	PNEC _{stat}	PNEC _{determ}
SiNC	µg SiNC per L	14 ^a	121	17–409	Accepted ^b	24	0.001
DCOIT	µg DCOIT per L	34 ^c	1	0.3–2	Accepted ^d	0.2	0.0004
Ag	µg Ag ⁺ per L	46 ^c	9	5–15	Accepted	2	0.006
SiNC–DCOIT	µg DCOIT per L	12 ^a	26	3–100	Accepted	5	0.007
	µg SiNC per L	12 ^a	116	13–446	Accepted	23	0.029
	µg SiNC–DCOIT per L	12 ^a	142	16–546	Accepted	28	0.036
SiNC–Ag	µg Ag ⁺ per L	11 ^a	24	4–69	Accepted ^e	5	0.017
	µg SiNC per L	11 ^a	144	23–419	Accepted ^e	29	0.109
	µg SiNC–Ag per L	11 ^a	187	30–543	Accepted ^e	37	0.100
SiNC–DCOIT–Ag	µg DCOIT per L	12 ^a	11	1–39	Accepted ^f	2	0.002
	µg Ag ⁺ per L	12 ^a	10	1–37	Accepted ^f	2	0.002
	µg SiNC per L	12 ^a	49	5–176	Accepted ^f	10	0.01
	µg SiNC–DCOIT–Ag per L	12 ^a	76	8–101	Accepted ^f	15	0.01

^a Only one EC₅₀ value available for each species. ^b Except for Kolmogorov–Smirnov and Cramér–von Mises tests for normality that rejected a 0.1 significance level. ^c Only one EC₅₀ value available for several species, thus the calculation of the geometric mean was not possible in these cases. ^d Except for the Kolmogorov–Smirnov test where significance levels were 0.025 and 0.01 and for the Anderson–Darling test that rejected a 0.1 significance level. ^e Except for significance level 0.1 in Anderson–Darling, Kolmogorov–Smirnov and Cramér–von Mises tests. ^f Except for Anderson–Darling and Cramér–von Mises tests that rejected a 0.1 significance level and for Kolmogorov–Smirnov that rejected 0.1 and 0.05 significance levels.

in the BPR assessment report (based on the NOEC data).³⁹ On the other hand, contrary to what was observed for the HC₅ value, the estimated PNEC value for pyridine triphenylborane (PTPB) was 75 times higher than the estimated value for DCOIT (Table 1). In this comparison, DCOIT appears to be more hazardous. It was expected that target species (*e.g.* *V. fischeri*, *S. costatum*, *M. galloprovincialis*, *M. edulis* and *A. amphitrite*) would be more sensitive towards DCOIT and their positions on the respective SSD curves were relatively lower than non-target species, such as microalgae, crustaceans or fish. DCOIT affects non-target organisms at lower concentrations compared to fouler species located in the base of the trophic chain (*e.g.* microalgae, copepods, rotifers and macroalgae) or at higher trophic levels (*e.g.* echinoderms and fish). This confirms the well-known broad-spectrum action of DCOIT posing a high risk to the ecosystem. This potential impairment of low and high levels of the trophic chain can represent a potential disruption of ecosystem functioning. Besides interfering with other metabolic pathways, DCOIT inhibits glutathione reductase by irreversible binding to the enzyme active loci which may explain its marine hazard.^{3,40,41}

Silver is known by its antibacterial activity, so it was expected that the most sensitive species were bacteria, such as *V. fischeri* and *Synechococcus* sp., however this was not observed. This result shows that silver appears to have a broader spectrum of action against marine species besides its bactericidal activity. It seems to be even more efficient against non-microbial species, presenting potential hazard to marine ecosystems. A value of HC₅ 18 times lower than the obtained in the present study (with the L/E/IC₅₀ data) was found in the literature for marine species (Table 1). This difference may be due to the difference in the amount of data

used to derive the SSD in both studies, since in the present study data for 28 additional species were included.⁴² The Canadian Council of Ministers of the Environment⁴² also estimated a SSD for freshwater species and obtained a HC₅ value 1.13 times lower, suggesting that those species are more susceptible to silver than marine species. In fact, silver is already used in a wide-range of biocidal applications, such as domestic water filters, silver-coated ceramic filters or as a topical antiseptic agent, due to its high anti-biofilm activity.^{25,26,43} Moreover, although primarily known by its action against microorganisms, previous studies demonstrated that silver has deleterious effects on other organisms at very low concentrations,^{9,39,41,44–46} including marine foulers.^{21,43} Nevertheless, silver is not yet in the list of active substances of the category PT21 (anti-fouling products) under the European Biocidal Products Regulation (EU Regulation 528/2012).

The PNEC data for DCOIT and silver indicate the need to develop alternative high performance and less hazardous solutions for their incorporation in coatings. Nanotechnology has been proposed as a means to contribute for this paradigm change in the additive industry, specifically to prevent and control corrosion^{47–50} and biofouling.^{17–21} Recent studies highlighted the decrease of toxicity caused by the immobilization or encapsulation of anti-fouling biocides in engineered nanomaterials^{17–21} without compromising the antifouling efficacy against target or fouler species.^{17,19,21} The bacteria *V. fischeri* and the diatom *P. tricornutum* were among the most sensitive species to SiNC–DCOIT and SiNC–DCOIT–Ag, and the bacteria species is also sensitive to SiNC–Ag, demonstrating the anti-microfouling efficacy of these nanostructured biocides. The other tested target species, the mussel *M.*

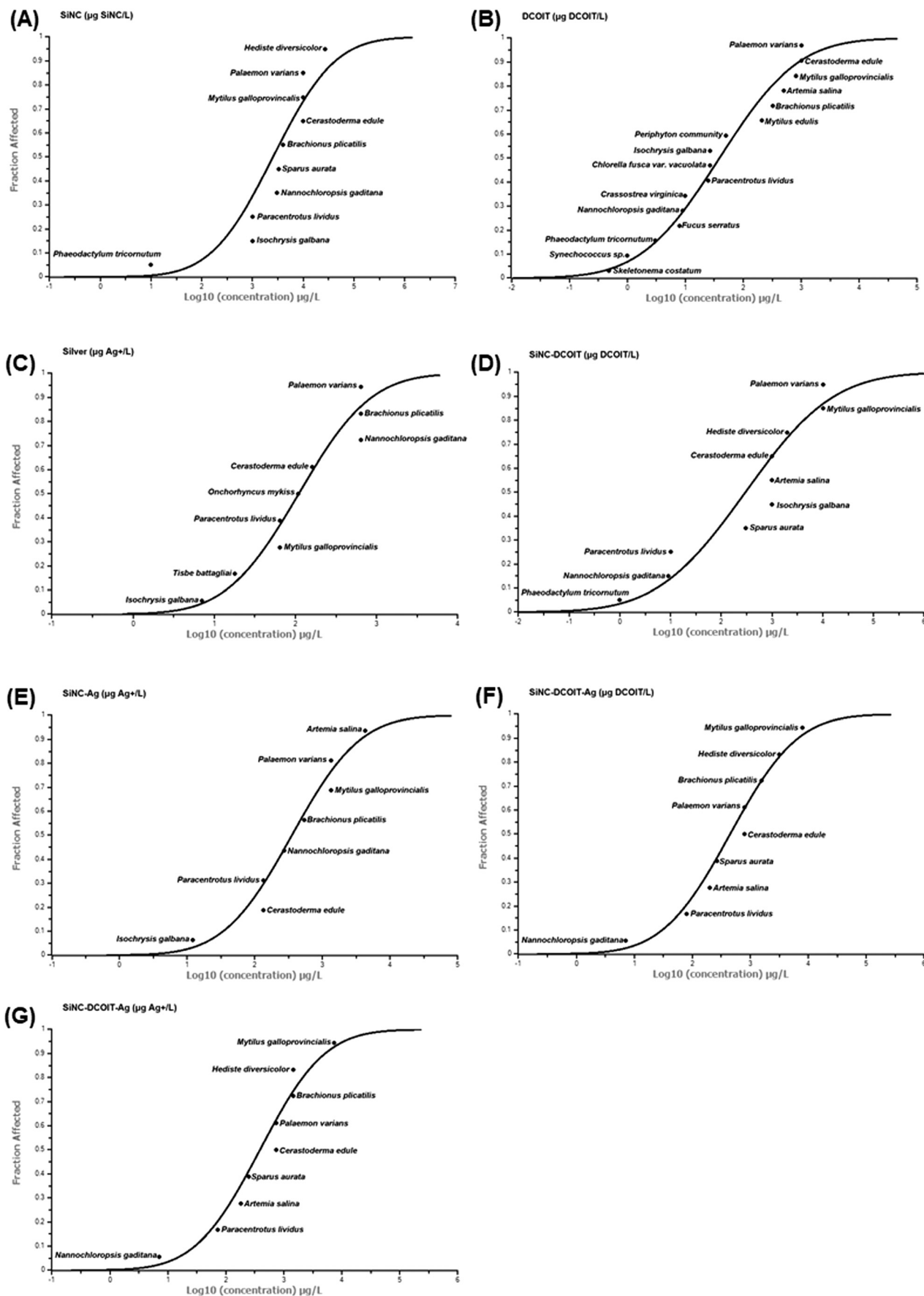


Fig. 2 Species sensitivity distributions for: (A) SiNC; (B) DCOIT; (C) Ag^+ ; (D) SiNC-DCOIT (as DCOIT content); (E) SiNC-Ag (as Ag^+ content); (F) SiNC-DCOIT-Ag (as DCOIT content); (G) SiNC-DCOIT-Ag (as Ag^+ content) using the NOEC data for different species.

Table 4 Hazardous concentrations 5% (HC₅) obtained from species sensitivity distributions using the NOEC data for the different species and PNEC values. *N* is the sample size and the tests for normality included the Anderson–Darling test, the Kolmogorov–Smirnov test and the Cramér–von Mises test. PNEC_{stat} – derived from the HC₅ with AF = 5; PNEC_{determ} – derived from the lowest L/E/IC₅₀ with AF = 1000 (according to TGD (2003))

Contaminant	Units	<i>N</i>	HC ₅	CI 95%	Normality tests	PNEC _{stat}	PNEC _{determ}
SiNC	µg SiNC per L	10 ^a	59	4–267	Accepted ^b	12	0.1
DCOIT	µg DCOIT per L	16 ^c	1	0.1–2	Accepted	0.2	0.0048
Ag	µg Ag ⁺ per L	9 ^c	7	1–22	Accepted	1	0.007
SiNC–DCOIT	µg DCOIT per L	10 ^a	1	0.03–12	Accepted ^d	0.2	0.01
	µg SiNC per L	10 ^a	4	0.1–54	Accepted ^d	1	0.04
	µg SiNC–DCOIT per L	10 ^a	5	0.1–66	Accepted ^d	1	0.1
SiNC–Ag	µg Ag ⁺ per L	8 ^a	15	1–61	Accepted	3	0.12
	µg SiNC per L	8 ^a	89	6–359	Accepted	18	0.7
	µg SiNC–Ag per L	8 ^a	114	8–458	Accepted	23	1
SiNC–DCOIT–Ag	µg DCOIT per L	9 ^a	12	1–56	Accepted	2	0.07
	µg Ag ⁺ per L	9 ^a	12	1–52	Accepted	2	0.07
	µg SiNC per L	9 ^a	56	5–251	Accepted	11	0.32
	µg SiNC–DCOIT–Ag per L	9 ^a	87	8–389	Accepted	17	0.5

^a Only one EC₅₀ value available for each species, thus the calculation of the geometric mean was not possible. ^b Except for the Anderson–Darling test that rejected 0.05 and 0.1 significance levels and for Cramér–von Mises that rejected a 0.1 significance level. ^c With the exception of *Mytilus galloprovincialis* and *Paracentrotus lividus* for DCOIT and *Tisbe batagliai* for Ag, only one EC₅₀ value is available for each species, thus the calculation of the geometric mean was not possible in those cases. ^d Except for the Kolmogorov–Smirnov test that rejected a 0.1 significance level.

galloprovincialis, exhibited an apparent tolerance to the abovementioned biocides as well as their free forms, which may be explained by their biological mechanism of closing their valves to avoid contamination.²¹

The deterministic approach, the most conservative one to determine the PNEC values in this study, highlighted that the encapsulation of DCOIT decreased at least 2 (SiNC–DCOIT) and 5-fold (SiNC–DCOIT–Ag) the DCOIT hazard, respectively. Compared to its free form, the hazard of silver was reduced by one order of magnitude in both SiNC–Ag but was not reduced in SiNC–DCOIT–Ag, using the NOEC data. The combination of both biocides in a nanostructured form proves to be more effective against target bacteria and diatoms, here used as a proxy of microfoulers, and simultaneously less toxic towards the tested non-target organisms compared with the free biocides, as seen in the SSDs and the estimated PNEC values. The lower hazard of these novel nanomaterials may be explained by their slow biocidal release capacity with time and environmental behavior in seawater, particularly, aggregation/agglomeration with time and variability with the exposure concentration.²¹ However, for a full comprehension of the hazard patterns here observed, we cannot discard the effects of the encapsulation on the stability and degradation of the organic biocide, among other aspects not addressed in the present study.

This achievement indicates that the encapsulation seems to be a promising technique to reduce the ecological hazard of DCOIT and, to a lesser extent, Ag. These findings also emphasize the importance of using both approaches to derive PNEC values and to obtain complementary insight aimed at ensuring appropriate environmental protection. Additionally, chronic testing is recommended for nanostructured substances, particularly on crustaceans and

fishes, in order to increase the knowledge on their long-term effects on marine organisms and refine the obtained PNEC values for these substances.

5. Conclusions

The present study provides the first holistic hazard assessment based on short-term toxicity data of new antifouling nano-based additives and it includes an extensive compilation of toxicity data for all tested substances that can be useful in a regulatory context, but also for industry.

The encapsulation of DCOIT and Ag in SiNC seems to be a promising eco-friendly solution for the development of innovative additives for anti-fouling coatings with lower hazard than the current commercial booster biocide. In the most conservative approach for the biocide hazard assessment, DCOIT hazard decreased at least 2-fold when encapsulated in SiNC (SiNC–DCOIT) and 5-fold when encapsulated together with silver (SiNC–DCOIT–Ag). The hazard of silver was also reduced by at least one order of magnitude in both SiNC–Ag and SiNC–DCOIT–Ag, in the most conservative scenario (deterministic approach using NOEC data).

Further studies are needed to address the nanomaterial hazard based on long-term effects, and to understand sub-cellular and bioaccumulation effects.

Conflicts of interest

No conflict of interests are declared.

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